

AI-Agent Systems for the CMS Analysis Workflow

Hyeon-Seo Yun, Norbert Neumeister, Dmitry Kondratyev, Ram Krishna Sharma

August 28, 2026

Introduction

- Goal of this talk: present a my experience in using LLM-based agent to accelerate my workflow.
- Outline:
  - My background
  - What an “AI-agent system” means in this context
  - Motivation: shortening the “time to insight”
  - Implementation with LLMs, and the minimal three-agent setup
  - Two worked examples and summary

My background

- My name is Hyeon-Seo Yun, 7th year PhD student
- Work on Run3 $H \to \mu\mu$ analysis under Prof Neumeister with Ram and Xudong.
- Got inspiration on from Arghya's preliminary talk about AI agent system.
  - Unfortunately, he ghosted me after our first date :/
- Did a minimal internet search on "AI-agent system" and jogged down the results over this Summer
- Share my work with others in Purdue CMS and set a simple baseline.
- I would like to thank Dmitry for supporting Purdue Analysis Facility (PAF) to be compatible with my AI-agent setup.
- **Recreatibility is a cornerstone of science**. Don't hesitate to visit me in my office Room 377 if your experience differs.

AI-agent systems: definition

- **AI-agent system** is a set of LLM agents whose behaviour is defined by protocols written down in a `SKILL.md` file for each agent.
  - Each agent has its own `SKILL.md`: what it is responsible for, what it may do, what it must hand off.
  - The protocol is plain text $\rightarrow$ human-readable, version-controlled and reviewable.
- These agents can be set into a workflow "loop" to keep iterating on a feature or a study.
  - Another name: "loop engineering".

Motivation: shortening "time to insight"

- Why is AI-agent system worth talking about? Because it helps to shorten **time to insight** and gain productivity.
  - **David Walter** from W mass measurement group dedicated a full slide about this concept in his seminar back in Nov 2025.
  - CEO of BOOM supersonic: faster iteration cycle allowed his team to make:
    - "Counterintuitive but much better design" for their supersonic jet
    - Workforce of 50 (vs traditional 500-5,000).
- The work I will be presenting today has significantly improved my time to insight for my personal work.
- The nature of the improvement is **not** Analysis specific.

Previous CMS work in shortening time to insight

- Time to insight is nothing new to CMS.
- Other CMS projects have been underway for years to improve automation of the CMS workflow. For instance:
  - **PocketCoffea**, where the full skimming workflow (trigger, object selection, ...) is defined over YAML config files.
    - **Zero** changes in the Python source code are necessary.
  - **Our Higgs to MuMu bash script** : Spearheaded by **Ram**, we have a wrapper layer that consolidates all the Analysis steps into one single line command.
    - Very efficient if you need to run many iterations of the same essential task.
- $\rightarrow$ shortens time spent coding the framework $\rightarrow$ accelerates the development cycle $\rightarrow$ shortens the time to insight.
- AI agent system is a **evolution** of what CMS has been already working on.

Implementation with LLMs: where we are today

- Many are already been using LLMs to generate code, but **validation and debugging** were left to the physicists (us) to still do.
- Example loop (old):
  - Physicist asks LLM to add a small new feature A
  - $\rightarrow$ LLM generates feature A
  - $\rightarrow$ Physicist tests feature A, finds a bug, and tells the LLM the bug **and** how to fix it.
- Physicst's time and mental capability during a work day is the biggest bottleneck.

Implementation with LLMs: AI agent system

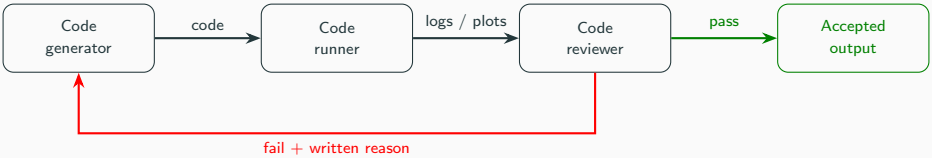
- Offload more work to LLMs
- Advances in LLMs (better common sense) allows the superficial / repetitive portion of validation and debugging work to be allocated to the LLM.
- Example loop:
  - Physicist asks LLM to add a small new feature A **and** to add basic validation tests that must pass
  - $\rightarrow$ LLM generates feature A
  - $\rightarrow$ LLM tests feature A, finds the bug
  - $\rightarrow$ LLM fixes the bug, repeats the test until it passes
  - $\rightarrow$ Physicist takes a final look.
- The physicist moves from **inside** the loop to **at the end** of the loop.

Setting up the AI-agent system

- I have set up a **minimal** AI-agent system with three agents:
  - **Code generator**: writes the code to run.
  - **Code runner**: executes the written code, applies parallelization if available.
  - **Code reviewer**: inspects the output from the code runner. If acceptable, end the loop. If not acceptable, leave the reason why it is not acceptable and pass the work back to the code generator.
- Demo repository with instructions on how to set it up (QR below):
  github.com/green-cabbage/Aug27_2026_AiAgentSystem_demo



Setting up the AI-agent system: the loop



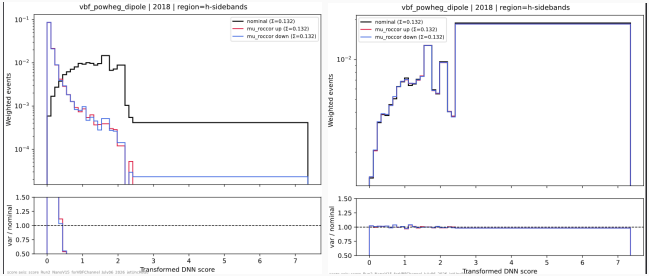
- The reviewer never edits code; it only accepts or returns work with a written reason.
- The physicist inspects the accepted output and the reasons recorded along the way.

Task 1

- Starting with a **simple** task: debugging.
- Bug in my MC template fitting where the up/down variations were very different for one specific systematic uncertainty.
- Fairly confident of the root source of the bug, but wanted to see if AI agent system could solve it.
- Main conditions:
  - Code Runner: Use scripts that I already use to generate results.
  - Code Reviewer: Delta between up/down templates and nominal template should be similar to other systematic variations (unless specified). If one variation is very different to others, raise a red flag.
  - Stop after N iterations and write report.
- Here's the link to the prompt text I gave to my LLM chatbot: taskPrompt

Task 1: Results

- The AI agent system flagged weird looking results, looked at the logs, found the bug and solved it.
  - NOTE: the code that printed the key output logs were manually written by me long time back. AI-agent didn't print them by themselves.
- Please see the results below for MC template comparison of the old buggy result (LEFT) and the new fixed results (RIGHT).



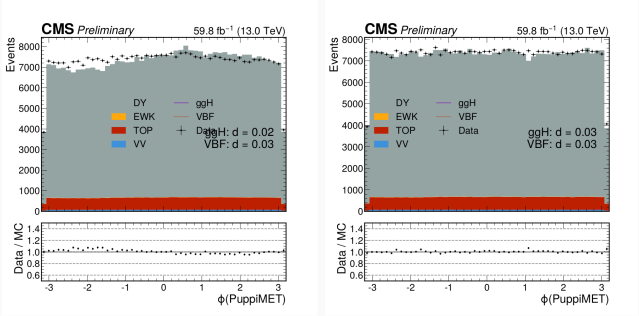
- Time spent on this task:
  - Human thinking and writing the setup: 2 hours
  - Time AI agent system spent time finding the bug and correct it: < 2 hours
  - Time spent on human validation: < 1 hour

Task 2

- Adding more complexity: adding new feature.
- Add MET xy correction scheme such that MET ϕ data/MC agreement is improved.
- Main conditions:
  - High-level descripton of the task.
  - **Code Generator**: Read CMS recommended recipe and other materials saved locally in `ref_material/MET_phi_corr`.
  - **Code Reviewer**: See that flatness of data/MC histogram is improved from reference.
- Here's the link to the prompt text I gave to my LLM chatbot: taskPrompt

Task 2: Results

- Surprisingly the AI-agent system shot back, saying that my instructions not consistent.
- The reference material that I have provided were not consistent with the task of MET ϕ.
- I found my mistake, gave the correct reference material and the task was executed and then validated.
- Please see below the MET ϕ control plots before (LEFT) and after (RIGHT) the implemented correction.



- Time spent on this task: Total 5 hours, with 3 hours of it being human work.

Summary

- **Task 1**: Successful debugging of an issue that required the human expert to think for few days on the side.
  - Had access to many output prints and logs to parse over.
  - Successful demo of AI agent system doing a simple task, but tedious for physicists.
  - LLMs are great at parsing over tons of log files, humans are terrible.
- **Task 2**: Successful implementation of a new object correction, with capability to push back on wrong setup.
  - The actual coding was simpler than Task 1, since it essentailly copied of reference code.
  - However, successful demo of AI agent system capable of reading correct material for correct implementation.
- Portions that need more work:
  - Capability of automating more complex tasks.
  - CMS specific conventions that make AI agent system report transparent and appriopriate.
- **Conclusion**: Over this summer, AI-agent system significantly increased my productivity. However, as a collaboration, we need to develop a common protocol framework that produces AI agent system specifically for CMS.

Small follow-up: reading an AN using Claude Opus 5

- Methodology:
  - Manually read the introduction, conclusion and figures, and understand them at a high level.
  - Build a mental map of the analysis, even if the technical details are filled in with my own personal assumptions.
  - Ask Claude to read the AN and make a high-level summary.
  - Read the summary; ask questions about the parts I would like to know more about or that I disagree with.
  - Repeat.
- Key points:
  - It is much easier on the brain to **verify** Claude's claim than to come up with the claim yourself.
  - More useful when you are tired, or when the wording in the AN is not very clear.

Backup

Restricting the phase space

- There is a lot of hype in “AI” (some justified and some not) so I would like to clearly draw a line.
- Clarify which parts of the CMS workflow are subject to change in this AI-agent system, and which are not.
- The dimension subject to automation is specifically the **AN workflow**.
  - Any layer more **fundamental** than it (physics objects, calibrations, POG recommendations, the framework itself) is **not** subject to change.
  - Any **higher-level** portion is subject to significant change.

